📓 SOC Internship – Day 12 Notes

🔍 Detection Rules & Use Case Development (Advanced)

1. ◇ What is a Detection Rule?

A **detection rule** is a logical condition used in SIEM tools to identify suspicious or malicious activities from logs.

👉 It helps SOC analysts automatically detect threats.

Example:

- IF failed login attempts > 5 from the same IP within 2 minutes
- THEN trigger alert: *Possible Brute Force Attack*

2. ◇ What is a Use Case?

A **use case** is a structured scenario that defines:

- What to monitor
- How to detect it
- What action to take

👉 It is more detailed than a detection rule and includes the full detection process.

3. ◇ Example Use Case – Brute Force Attack

- **Use Case Name:** Brute Force Detection
- **Data Source:** Authentication Logs
- **Condition:** Multiple failed login attempts from same IP
- **Detection Logic:** More than 5 failures in a short time
- **Response:**
  - Generate alert
  - Block IP address

4. ◇ Mapping to MITRE ATT&CK

Detection rules should be mapped to the **MITRE ATT&CK framework** for better threat understanding:

- **Brute Force → Credential Access**
- **Phishing → Initial Access**
- **Malware Execution → Execution**

5. ◇ Best Practices for Detection

✓ Avoid false positives (reduce unnecessary alerts)
✓ Use clear and simple conditions
✓ Continuously improve detection rules
✓ Always map detections to MITRE ATT&CK